Description of files in this folder

1. run_trim_galore.sh
    a. Command used to run TrimGalore on the samples
    b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run_kallisto.sh
    a. Command used to run kallisto after trimming
3. sample_info_mouse_brain_rnaseq.xlsx
    a. Metadata used for the analyses below
4. pca_and_deseq2.Rmd
    a. Inputs:
        i. The Kallisto output files which are available on GEO (GSE341280)
        ii. The t2g.txt that came with the standard mouse transcriptome index that was used by kallisto for pseudoalignment
            1. Downloaded from the following link on March 5, 2025:
                a. https://github.com/pachterlab/kallisto-transcriptome-indices/releases/download/v1/mouse_index_standard.tar.xz
        iii. sample_info_mouse_brain_rnaseq.xlsx (metadata, available in this folder, see above)
    b. Outputs:
        i. PCAs for each mouse brain RNA-seq experiment (there were two, see Methods) – used for a supplementary figure
            1. expt1_samples_pca.pdf
            2. expt2_samples_pca.pdf
        ii. hemoglobin_genes_plot.pdf – used for a supplementary figure
        iii. cooks_distance_tmao_samples.pdf – used for a supplementary figure
        iv. DESeq2 results, used to create a supplementary table
            1. Contained within deseq2_outputs subfolder
        v. Volcano plots for each metabolite tested (TMAO, PA, IAA, GDCA) – used for a main figure
5. fgsea.Rmd
    a. Inputs:
        i. The DESeq2 results (produced by the previous script) for TMAO and GDCA
        ii. The following collections of gene sets, downloaded on July 4, 2025 from: https://www.gsea-msigdb.org/gsea/msigdb/mouse/collections.jsp
            1. Hallmark gene sets – “mh.all.v2025.1.Mm.symbols.gmt”

            2. REACTOME subset of CP (canonical pathways) – "m2.cp.reactome.v2025.1.Mm.symbols.gmt"
    b. Outputs:
        i. All contained within the fgsea_results subfolder:
            1. Results of gene set enrichment analysis, using the input collections of gene sets
                a. Used for a supplementary table
            2. PDFs depicting selected significant terms from GSEA
                a. Used for figures
6. marker_gene_analysis_tmao.Rmd
    a. Inputs:
        i. Processed single-cell data from https://doi.org/10.1038/s41593-019-0491-3, downloaded from https://singlecell.broadinstitute.org/single_cell/study/SCP263/aging-mouse-brain
        ii. Genes significantly regulated by TMAO, as outputted by the DESeq2 script above
    b. Outputs:
        i. Plot showing % of marker genes downregulated by TMAO, for each of six brain cell classes
            1. Used for a main figure
        ii. RDS containing the computed marker genes per class
        iii. TSV containing the computed marker genes per class, used for a supplementary table
7. marker_gene_analysis_gdca.Rmd
    a. Inputs:
        i. The RDS outputted by the previous script
        ii. Genes significantly regulated by GDCA, as outputted by the DESeq2 script above
    b. Outputs:
        i. Plot showing % of marker genes downregulated by GDCA, for each of six brain cell classes
            1. Used for a main figure
8. gaba_a_receptor_genes_heatmap.Rmd
    a. Inputs:
        i. Same as those for pca_and_deseq2.Rmd, above
    b. Outputs:
        i. A heatmap, used for a supplementary figure, showing downregulation of $GABA_A$ receptor subunits in the brains of TMAO-treated mice

9. star_alignment_script.sh
    a. Bash script that was used to run STAR alignment (instead of kallisto) on the trimmed fastq files, in order to prepare for differential splicing analysis with rMATS-turbo
10. star_index_script.sh
    a. Bash script that was used to create the STAR index for STAR alignment
11. rmats_turbo_script.sh
    a. Bash script that was used to run rMATS-turbo after STAR alignment
12. differential_splicing_events.Rmd
    a. Inputs:
        i. Files outputted by rMATS-turbo, available in this folder within the subfolder rmats_output
    b. Outputs:
        i. Excel files
            1. diff_splicing_analysis_results.xlsx
            2. sig_diff_splicing_events.xlsx (used for a supplementary table)
        ii. Volcano plot used for a main figure panel
13. sashimi_code subfolder
    a. Contains files used to generate supplementary sashimi plots of differential splicing in *Rsrp1*, *Cpsf4*, *Ythdf3*, and *Kmt2b* in brains from TMAO- vs. saline-treated mice.
    b. Specifically contains:
        i. Files containing the commands used to run ggsashimi.py (which was downloaded from https://github.com/guigolab/ggsashimi on June 26, 2025)
        ii. Input files required for running those commands:
            1. palette_saline_grey.txt
            2. input_bams.tsv
        iii. Raw output PDFs of the ggsashimi commands, which were later edited in Illustrator to reduce visual clutter and emphasize the junctions relevant to each differential splicing event
    c. Required files not included in this repo:
        i. Bam files (generated by STAR – very large)
        ii. gencode.vM37.primary_assembly.sorted.gtf
            1. This file was generated from gencode.vM37.primary_assembly.annotation.gtf, which was downloaded from GENCODE (release M37; GRCm39; https://www.gencodegenes.org/mouse/).

2. The annotation file was sorted on the command line using “sort -k1,1 -k4,4n”